

Closed-form 2nd-order Data Shapley as a Spurious-Correlation Attenuator

Where it works, where it doesn't — and why

A closed-form, single-pass per-sample reweighter for worst-group robustness.

Mechanism reproduced across **three** spurious regimes: color, texture, natural images.

The problem: spurious correlations break worst-group accuracy

- Models latch onto an **easy spurious cue** (background, color, texture) that correlates with the label in training but **flips at test time**.
- Average accuracy looks fine; the **worst group collapses**.
- Strong fixes need **group labels** (GroupDRO) or expensive two-stage/bi-level pipelines (JTT, meta-reweighting).

Question: can a **closed-form, single-pass** per-sample reweighter — no group labels at training, no inner optimization loop — attenuate the spurious cue?

Our answer

Repurpose **In-Run Data Shapley** (Wang et al. 2024) as an in-training reweighter, and add a **2nd-order gradient-Hessian-gradient cross-term**. We isolate exactly what the cross-term contributes and show it generalizes across three spurious modalities.

Idea: Data Shapley as an in-training reweighter

- In-Run Data Shapley gives a **closed-form Taylor estimate** of how much each training sample improves a small balanced validation anchor — computed **during** training, no retraining.
- We turn that valuation into a **per-sample weight** and run weighted SGD.
- A 50:50 group-balanced **anchor** is used only for the inner inner-product — sensitive labels are never consumed by the model update.

$$w_i = B \cdot \text{softmax}(\text{EMA}(\phi_i) / \tau) \rightarrow \theta_{t+1} = \theta_t - \eta \sum_i w_i \nabla L(z_i)$$

The whole question becomes: **what should ϕ_i be?**

Method — closed-form Shapley score (1st & 2nd order)

$$\varphi_i^{(1)} = \langle g_i, g_{\text{val}} \rangle \rightarrow \varphi_i^{(2)} = \varphi_i^{(1)} - \alpha \cdot \langle g_i, H_{\text{val}} g_{\text{val}} \rangle$$

- **1st order** (Fairds-1): $\varphi^{(1)}$ = how much z_i reduces validation loss in one SGD step. Per-sample gradients via `torch.func.vmap`, single graph — closed form, no bi-level loop.
 - **2nd order** (Fairds-2): add the gradient–Hessian–gradient cross-term via the **Pearlmutter trick** (one extra backward pass, no full Hessian).
 - **Algorithmic contribution**: rescale the cross-term to the **same RMS magnitude** as $\varphi^{(1)}$ each minibatch — H_{val} 's spectrum varies by orders of magnitude.
- ⚠ **Without RMS rescaling, Fairds-2 collapses to chance under strong spurious bias.**

Is the 2nd-order term doing something real?

A skeptic's worry: maybe the cross-term is just **smoothing / regularization** noise. We tested this directly.

What we found about the geometry

- g_{val} is almost the **top eigenvector of H_{val}** (alignment 0.89–0.99) $\Rightarrow$ the cross-term is **nearly parallel** to $\phi^{(1)}$.
- Decompose: $\text{cross}_n = \beta \cdot \phi^{(1)}$ (parallel, a curvature shrinkage) + $\mathbf{r}$ (orthogonal residual).

The parallel part = global shrinkage (equivalent to smoothing).

Is the orthogonal residual $\mathbf{r}$ a real, sample-specific signal — or noise?

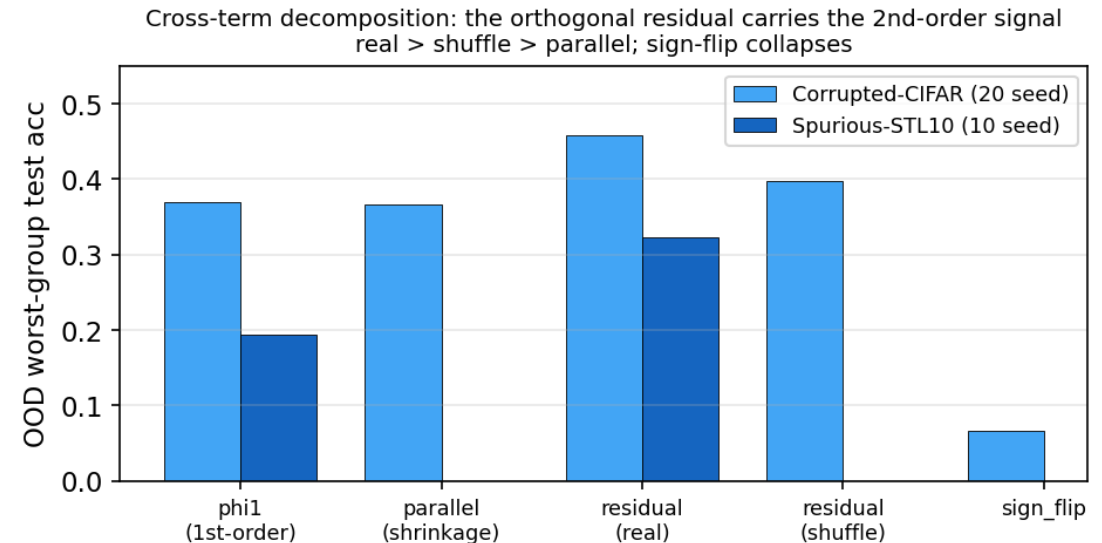
Residual ablation: the cross-term's signal lives in r

We build 5 arms that share $\varphi^{(1)}$ but treat r differently:

- **residual (real):** $\varphi^{(1)} - \alpha \cdot r$
- **residual (shuffle):** r permuted within class $\times\varphi$ -bin
- **parallel:** only the shrinkage part
- **sign_flip:** $\varphi^{(1)} + \alpha \cdot r$

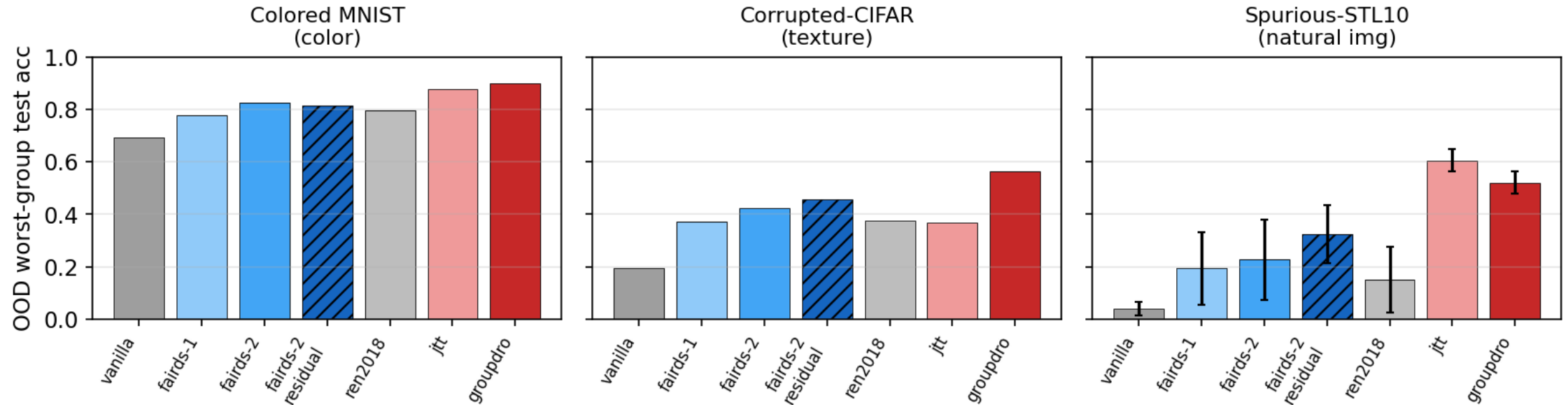
real > shuffle > parallel;
sign-flip collapses.

CIFAR: real vs shuffle **+5.9pp**, **p=0.0015** (20 seeds). If r were noise, shuffling it would not matter — it does.



Headline: the mechanism reproduces across 3 spurious regimes

Fairds-2 (residual) beats no-group-label baselines across 3 spurious regimes (grey/blue = no group label; red = uses group label or 2-stage)



Fairds-2 (residual) beats every **no-group-label** baseline (vanilla, Fairds-1, Ren2018) in **all three** regimes — color (MNIST), texture (CIFAR), and natural images (STL-10).

The third regime (STL-10, 96px natural images)

New benchmark: car vs truck, corruption-type (blur vs noise) as the spurious cue; from-scratch CNN, 10 seeds.

method	OOD test_worst	vs Fairds-2-resid
vanilla	0.042	+28.1pp, p=0.002
Fairds-1 (1st-order)	0.194	+12.9pp, p=0.010
Ren2018 (bi-level)	0.151	+17.2pp, p=0.002
Fairds-2 (residual)	0.323	—
JTT / GroupDRO*	0.61 / 0.52	stronger (see next)

2nd-order isolation holds: the residual variant beats the 1st-order term by +12.9pp (p=0.010) — exactly the cross-term effect we isolated on CIFAR, now in a third modality.

Honest scope — what we do *not* claim

Stronger methods exist (reported plainly)

- **JTT** (2-stage ERM) and **GroupDRO** (uses group labels) reach higher worst-group accuracy in most regimes.
- Fairds-2's value is the **closed-form, single-pass cross-term mechanism**, not peak accuracy.

A regime where it fails: pretrained-ResNet Waterbirds fine-tuning

- All Fairds variants (3 algorithmic versions) **fail to match** the bi-level baseline.
- Cause: with a strong pretrained extractor, early per-sample gradients are small/noisy $\Rightarrow$ the EMA buffer fills with noise.

Use bi-level meta-learning there.

We characterize the **regime boundary** rather than hide the negative result.

Cost & practicality

Wall-clock overhead (ResNet-18)

method	×vanilla
Fairds-1	4.30×
Ren2018 (bi-level)	4.78×
Fairds-2	7.68×

1st-order is cheaper than bi-level; 2nd-order pays for the HVP. Honestly reported, not yet at the 1.5×/3× target.

When to reach for Fairds-2

- **From-scratch / small models** with a strong spurious cue and no group labels.
- When you want a **closed-form** reweighter (no inner loop) and lowest seed variance among no-label methods.
- **Not** for pretrained fine-tuning ⇒ use bi-level.

Contributions & takeaway

- Repurpose closed-form In-Run Data Shapley as an **in-training per-sample reweighter** for worst-group robustness.
- A **2nd-order cross-term with RMS rescaling** (required for stability under strong spurious bias).
- A **decomposition + residual ablation** proving the cross-term carries a real, sample-specific directional signal — real > shuffle > parallel; sign-flip collapses — **not mere smoothing**.
- The mechanism **reproduces across three regimes** (color / texture / natural images), with an honest negative on pretrained fine-tuning.

Closed-form valuation can shape training dynamics — within a characterized regime.

Beats every no-group-label baseline from scratch; group-label / 2-stage methods remain stronger, and we say so.